

Abhay Patel

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EXPERIENCE

Jan 2026 – Present	Kubrick Group – Data Engineer - Developed cloud-based data engineering solutions using Databricks, Apache Spark, Delta Lake, and Unity Catalog, building scalable ETL pipelines following production engineering practices. - Applied SQL, Python, and data modelling techniques to deliver reliable data solutions, analytics products, and cloud-based data pipelines. - Utilised Power BI, Git, Azure DevOps, and machine learning fundamentals to support collaborative development and data-driven decision making.
Oct 2023 – Dec 2023	Explore Learning – Academic Tutor Delivered tailored 1:1 and group sessions to students aged 5-16, including exam and SEN support
Oct 2017	UTC Aerospace Systems - Internship Assembled and tested fuel and hydromechanical units, analysing technical drawings and resolving discrepancies for Rolls-Royce engine components.

EDUCATION

Sept 2020 - Aug 2023	The University of Manchester BEng Aerospace Engineering with Honours
Sept 2011 - 2019	King Edward VI Five Ways Grammar School A level: Mathematics (A), Physics (A), Chemistry (A)

PROJECTS

Mar 2026 – April 2026	NYC Restaurant Delivery Service – Kubrick Group - Developed an end-to-end ELT pipeline using Databricks and Microsoft SQL Server, implementing a Medallion architecture and contributing to the silver layer through data cleaning, validation, and transformation to create reliable analytics-ready datasets. - Built Gold layer data models and Power BI dashboards tailored to three stakeholder groups, translating business requirements into actionable insights through data visualisation and reporting. - Delivered a sales forecasting analysis by aggregating restaurant delivery data to hourly intervals, evaluating the impact of external factors including sports events, weather, holidays, and seasonality; identified a ~30% increase in orders during major league game windows.
Feb 2025 – April 2025	Trading Strategy Quantitative Backtesting - Built a Python-based multi-stock backtesting engine using Pandas and NumPy, implementing trading strategies, technical indicators, risk controls, and performance metrics. - Developed a Streamlit dashboard to visualise strategy performance, equity curves, and asset behaviour.
Oct 2023 – Jan 2023	Earth Observation Satellite Design - Designed a lightweight vegetation-monitoring satellite structure in SOLIDWORKS, integrating payload, orbital, and launch vehicle constraints through cross-disciplinary collaboration. - Optimised alloy selection through material analysis, achieving a 25% reduction in material costs while maintaining performance requirements.
Oct 2022 – May 2023	Design of a Biologically-Inspired Flapping Wing Micro Aerial Vehicle - Developed MATLAB-based motor, aerodynamic, and dynamic models using regression-based scaling laws to evaluate 30+ configurations and optimise flapping-wing MAV performance, achieving system efficiencies of up to 41%. - Designed and prototyped carbon-membrane flapping wing structures in SOLIDWORKS, testing wing designs across varying frequencies to validate aerodynamic performance.

CERTIFICATIONS

July 2026 – 2028	Databricks Certified Data Engineer Associate
July 2026 - 2028	Databricks Certified Data Analyst Associate
June 2026	Dataiku Core Designer

ACHIEVEMENTS, SKILLS & INTERESTS

Computational Skills

- **Programming:** Python, SQL, MATLAB
- **Data Engineering:** ETL Pipelines, Data Modelling, Databricks, Apache Spark, Delta Lake, Unity Catalog
- **Cloud & Tools:** Azure, Git, Azure DevOps, VS Code
- **Analytics & ML:** Power BI, Scikit-Learn, Supervised/Unsupervised Learning

2023 – 2025

Solo Travel

- Independently travelled in 3 continents, building resilience, cultural awareness and confidence in unfamiliar environments

Nov 2018 – Jan 2019

London Institute of Banking & Finance - IFS Student Investor Challenge

- Ranked in the top 1% out of 4,000+ teams in a UK-wide virtual trading competition by managing a simulated investment portfolio and making strategic financial decisions

2012 – 2023

Competitive Sports

- University of Manchester Swim team, Kings Heath Cricket Club, Kings Heath Warriors Football Club

LEADERSHIP & ROLES OF RESPONSIBILITY

Sept 2022 – April 2023

University of Manchester Football Captain

- Led a 7-a-side team in university-campus tournaments, top 3 league finish.

Oct 2018 – 2019

RLSS National Lifeguard

- Maintained high safety standards and strengthened reassurance for swimmers.

Sept 2017 – 2019

King Edward VI Five Ways Grammar School Sixth Form Prefect

- Selected by senior leadership, oversaw main canteen operations for 1300+ students.